

W M+ Check Creator — End-to-End Physical Printing — Review Report

Branch: feature/windows-print-status-checkpoint · **Tip:** 783775c (2026-09-22, Rick) **Reviewer:** Jose (code-only review, no live test — physical printing runs on Windows) **Date:** 2026-09-23

Rick's test statement: fresh IPP readiness check → PDF with zero DB writes → physical submission through the Local Print Bridge → issuance only after successful submission → printer OFF blocks issuance → PRINTER NOT READY popup with retry/cancel → OFF→ON recovery → single + three-check sequential printing → checks 1026, 1027, 1028 verified Pending.

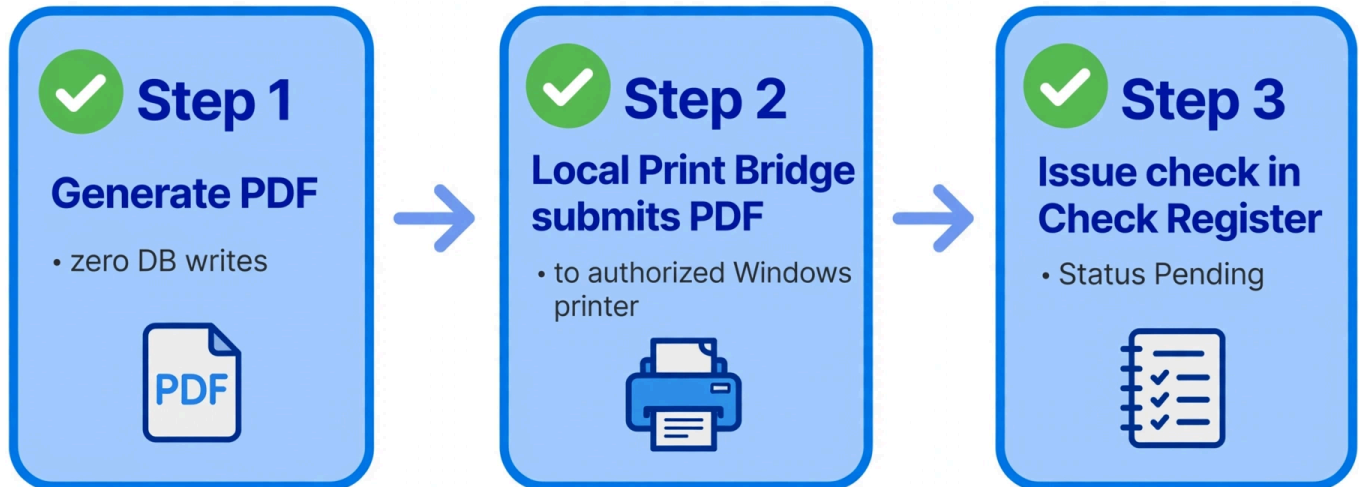
Commits under review

Commit	Title	Scope
ec2591b	WIP: Check Creator, PDFKit, MICR, and Check Register work	+2498/−44 — PDF engine, Check Creator UI, test-hack removal, Banking modes
f2be392	Windows IPP printer-status milestone	+350 — print bridge, printer discovery, PrintEngine
783775c	Complete end-to-end Check Creator physical printing	+511/−107 — pdf/issue split, /print-pdf , popups

Verification (code review vs. claimed behavior)

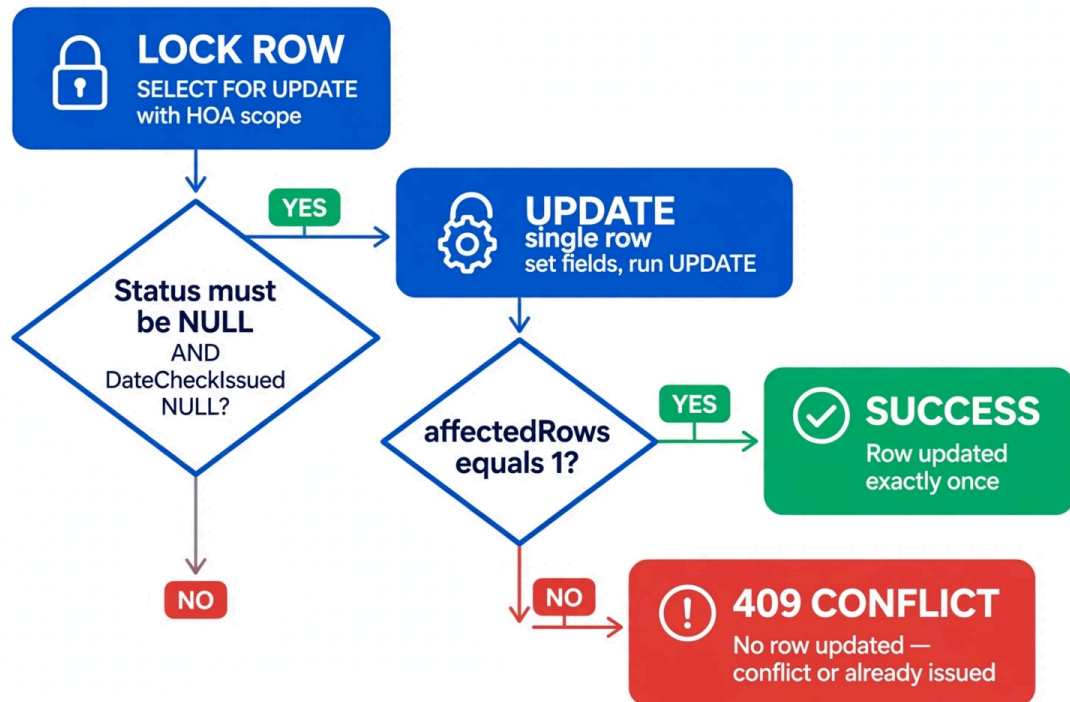
#	Claimed behavior	Code evidence	Result
1	PDF generation performs zero DB writes	New POST /api/print-checks/pdf uses pool db directly, no getConnection / beginTransaction ; loadCheckPrintData = SELECT + joins; getHoaTimeZone = SELECT SystemSettings + in-memory tz	☐ Confirmed structurally
2	Issuance only after successful bridge submission	Frontend loop per check: pdf → submitPdfForPrint → POST /api/print-checks/issue ; any throw skips issue	☐ Confirmed
3	Printer OFF prevents issuance	confirmPrinterReadyForPrint gate before loop; non-READY opens popup instead of throwing	☐ Confirmed
4	PRINTER NOT READY popup with retry/cancel	showPrinterUnavailable overlay: PRINTER IS ON re-checks + resumes, CANCEL PRINT closes	☐ Confirmed
5	Each printed check independently Pending	issue is per-transaction (FOR UPDATE + single-row UPDATE + commit per check)	☐ Confirmed
6	Completion confirmation + return to register	CHECK PRINTING COMPLETE popup with per-count message; OK calls onClose	☐ Confirmed
7	Race-safe issuance	SELECT ... FOR UPDATE with MgtCo+HOA scope; UPDATE requires DateCheckIssued IS NULL AND Status IS NULL ; affectedRows===1 else 409; button disabled via isCreatingChecks	☐ Confirmed
8	Test scaffolding removed	ec2591b deletes the TEMPORARY ALEX TESTING hack — new checks enter NULL/NULL (precondition for the guards)	☐ Confirmed
9	Bridge loopback-only	print-service.js listens on 127.0.0.1:3012 (before and after); PRINT_BRIDGE_URL is loopback; CORS * contained to the machine	☐ Confirmed

CHECK PRINTING SEQUENCE — PDF, Print, Issue



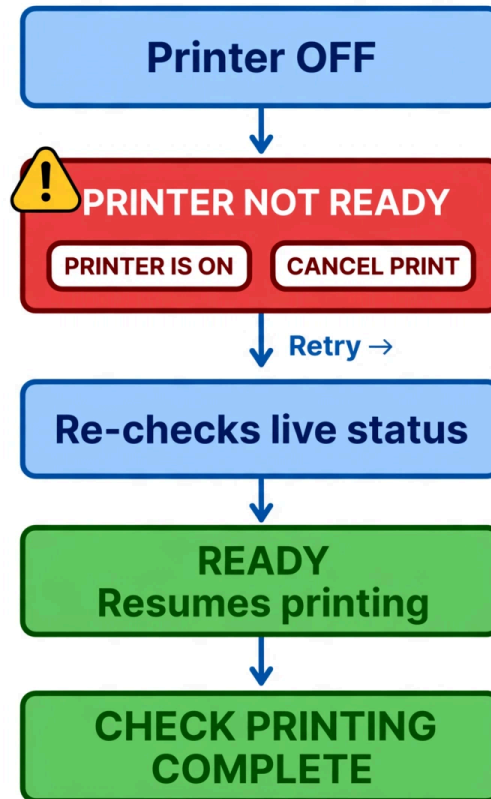
ISSUE GUARDS — One Row, Once.

Atomic guard to ensure only one row is updated once



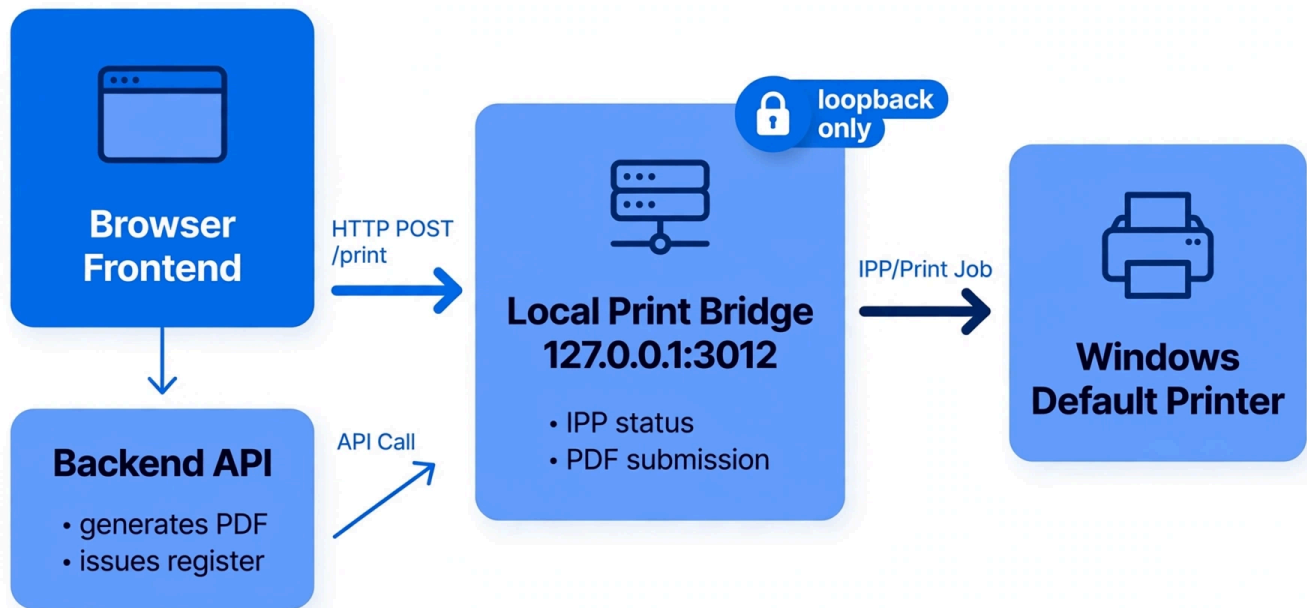
Guard ensures one-time update • Idempotent • Race-safe

PRINTER RECOVERY — Retry, Resume



PRINT BRIDGE ARCHITECTURE

Loopback Only



All traffic stays on localhost • 127.0.0.1 loopback interface only • No external network

Observations (non-blocking)

01 — Partial-batch resume reprints. If `issue` fails on check 2 of 3, retry calls `createSelectedChecks()` from check 1: the pdf endpoint has no Status guard, the bridge reprints physically, and only `issue` returns 409. Suggest a resume index (continue from the failed check) or a Status pre-check before bridge submission.

02 — No reprint path after a paper jam. Any existing `Status` now returns 409 and the UPDATE requires NULL/NULL (previously `Pending` could re-issue). Correct as "issue once", but post-issue jam recovery has no defined operator procedure.

03 — Bridge has no own package.json. `require('pdf-to-printer')` resolves via `backend/node_modules` (works only if backend deps are installed). Suggest an explicit `backend/print-service/package.json` or a startup check.

04 — Dead weight. Three empty placeholder components (`PrintChecksActions/Details/Lists.jsx`, 0 bytes) and binary blobs with no source in repo (`IppStatusHelper.exe`, 3 MICR `.ttf`, 3 `.dotx` templates) — accepted as-is, noted for provenance (native helper source requested).

05 — Pre-existing. Frontend `localhost:3011` URLs now in two endpoints; `pdf-preview` route untouched while the UI no longer opens PDFs (physical-only by design).

Auth integration

`POST /api/print-checks/issue` carries hardcoded scope with `T0D0(scope-merge)` markers (`HOA-FL-2024-001 / MGTC0-001 / SYSTEM`) — consistent with keeping `feature/auth-roles` separate until the integration design (security levels, active-HOA switching, session rules, frontend/cache clearing) is jointly agreed. The future merge will conflict exactly on these hunks; that is the right time to wire `req.hoa`.

Verdict

Approve with minor observations. Architecture is sound, race guards are correct, and the test-hack removal restores the production invariant the issuance flow depends on.

Open questions for Rick

1. Operator procedure for O1 (printed-but-not-issued) and O2 (post-issue jam)? 2. Confirm checks 1026-1028 started from NULL/NULL state? 3. `IppStatusHelper.exe` source or acceptance of the blob? 4. Do you want this branch merged to BravoFrontend, and with what strategy for the two local-only docs commits (`e0f7b3e` , `3ab74bf`)?